CHUNGMANN KIM

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Ellicott City, MD, USA

Research Interests

Global Food Security · Health & Nutrition · AI for Humanitarian · Climate Change · International Development · Poverty & Food Security Measurement

Current Academic Position

- **Bloomberg School of Public Health, Johns Hopkins University** — *2026 – Present*
  *Postdoctoral Research Fellow at the **Center for Humanitarian Health*** — United States

Education

- **University of Illinois at Urbana–Champaign** — *2025*
  Ph.D. in Applied Economics — United States
- **Seoul National University** — *2019*
  M.Sc. in Agricultural Technology — Republic of Korea
- **Handong Global University** — *2015*
  B.Sc. in Environmental Studies (Cum Laude Graduate) — Republic of Korea

Working Papers

"**Hearing the Alarm: Do Donors Follow Institutional Crisis Signals?**" [Draft]
Selected presentation at UNU-WIDER Development Conference (awarded full travel support) · European Association of Agricultural Economists (EAAE) 2025 Congress · Agricultural and Applied Economics Association (AAEA) 2025 Annual Meeting (Poster) · American Public Health Association (APHA) 2026 Annual Meeting and Expo (Oral Presentation)

"**Early-Year Milk Price and Child Stunting in Zambia**." [Draft | Slides]
Presented at AAEA 2022, EAAE 2023, Tata-Cornell 2023 · American Public Health Association (APHA) 2026 Annual Meeting and Expo (Poster).

"**H-Aid Tracker AI-Powered Subnational Humanitarian Aid Mapping**."
Presented at Food Assistance Strategic Dialogue, Johns Hopkins Bloomberg School of Public Health (Washington D.C., 2026) · AI + Health Care Idea Tank Pitch Finalist, Hopkins Business of Health Initiative, Data Science and AI Institute, & Innovation Translation Council (2026)

Publications (Peer-Reviewed)

"**Hidden Hunger: Global Estimates Systematically Undercount Acute Hunger**." Lentz, E., Michelson, H., Baylis, K., & **Kim, C.** (2025). *Nature Food* . [DOI] | [Replication Code]

"**Inside the Black Box: How Consistent Are Global Food Security Crisis Assessments**." Lentz, E., Michelson, H., Baylis, K., & **Kim, C.** (2025). *Food Policy* . [DOI] | [Replication Code]

"**Machine Learning for Food Security: Principles for Transparency and Usability**." Zhou, Y., Lentz, E., Michelson, H., **Kim, C.**, & Baylis, K. (2022). *Applied Economic Perspectives and Policy* . [DOI]

"**Exploring the Association between Agricultural Production Systems and Household Diets in Viet Nam**." Bandyopadhyay, A., Azzarri, C., Haile, B., **Kim, C.**, et al. (2022). *Food Security* . [DOI]

"**The Economics of the Soy Kit for Food Entrepreneurs**." **Kim, C.** & Goldsmith, P. (2021). *Food and Nutrition Bulletin* . [DOI]

"**Vietnamese Preferences for Fresh Korean Ginseng**." Joo, Y., Kim, T., Nguyen, D.K., & **Kim, C.** (2021). *Korean Journal of Agricultural Economics* . [DOI]

Work in Progress

"**Climate Change, Conflict, Displacement and Health in Mali and Iraq**." With Center for Humanitarian Health. Modeling displacement using AI and estimating climate-driven health needs. [Project Link]

"**Developing a Review and Evaluation Framework for AI Models (LACI Project)**." With Melissande, M. (EuropeanCommission – Joint Research Centre, EC–JRC) and IPC-GSU.

"**Acute Food Insecurity Data and Modelling**." With Ronco, M. et al. (collaboration with EC–JRC).

"**Multidimensional Food Security Indicators: Evidence from Sub-Saharan Africa**." With Baylis, K., Lentz, E., Michelson, H. Presented at AAEA 2022 Track Session.

Professional Experience

- **Food and Agriculture Organization (FAO), IPC Global Support Unit** — *Feb 2024 – Present*
  Data Scientist & Modelling Specialist — Rome, Italy / Remote
  ◦ Predictive modelling · Multi-Source Data Integration · AI Model Evaluation
  ◦ Prototype development: News Media Analytics for Risk Monitoring [Live Tool]
- **Food and Agriculture Organization (FAO), IPC Global Support Unit** — *Oct 2022 – Aug 2023*
  Food Security Analyst — Rome, Italy / Remote
  ◦ IPC Accuracy Study · Descriptive and inferential analytics
- **International Fund for Agricultural Development (IFAD)** — *Sep 2023 – Jan 2024*
  Impact Assessment Specialist — Rome, Italy / Remote
  ◦ IFAD11/12 dataset harmonization · 10+ countries · Geospatial and socio-economic indicators
  ◦ Create automated data pipelines · climate adaptation program evaluation (Nepal)
- **International Food Policy Research Institute (IFPRI)** — *Oct 2018 – Jan 2019*
  Research Intern — Washington, DC
  ◦ International Dietary Data Expansion (INDDEX) project, Vietnam
  ◦ Co-authored research on integrated farming systems and household diets
- **Canaan Farmer School** — *Jan 2015 – Dec 2017*
  Project and Training Officer — Korea & Tanzania
  ◦ Capacity building · Agriculture · Resilience · Microfinance
  ◦ Grant management · Cooperative development · Community development projects

Policy Work & Technical Projects

- **LACI Framework: Evaluating AI Models for IPC Use** — *2025*
  Policy Tool & Framework — FAO–IPC GSU
  ◦ Developed the Learning, Assessing, Calibrating, and Integrating (LACI) framework to evaluate AI, ML, and statistical model outputs for IPC integration
  ◦ Designed operational tools (Learn and Assess Cards) to conduct technical screening, define context-specific limitations, and map outputs to IPC workflows
  ◦ Applied the framework to validate early warning predictive models (e.g., World Bank's Yemen Data-Driven Model)
- **Optimizing Classification Thresholds for Food Security Indicators** — *2025*
  Technical Project — FAO–IPC GSU
  ◦ Threshold optimization · Household survey harmonization · Cross-country concordance
- **News Media Analytics for Risk Monitoring** — *2025*
  Policy Tool — FAO–IPC GSU & Kickstart AI (Netherland)
  ◦ News Monitoring using NLP / LLM · Sentiment analysis · Topic modelling · IPC risk integration [View Slides]
- **IPC Accuracy Study: Internal Consistency of AFI & AMN Analyses** — *2024*
  Technical Report — FAO–IPC GSU
  ◦ Food Security Measurement Study [View Report]
- **Production, Consumption, and Food Security in Viet Nam** — *2021*
  Diagnostic Report — IFPRI, FAO, Tufts Univ.
  ◦ Agricultural production · Diets · Food security linkages [View Report]
- **Mapping of Poultry Hotspots in Tanzania** — *2018*
  Technical Note — IFPRI
  ◦ Geospatial analysis · Market hotspots · Poultry [View Report]

Teaching

- **University of Illinois at Urbana–Champaign** — *Fall 2024*
  Teaching Assistant, ACE 251: The World Food Economy — Champaign, IL
  ◦ Delivered weekly in-class lectures: global food production, consumption, and trade · problems of hunger and population · the role of agricultural development, trade, and aid in relieving hunger
- **University of Illinois at Urbana–Champaign** — *Fall 2022, Fall 2023*
  Teaching Assistant, ACE 435: Global Agribusiness Management — Champaign, IL
  ◦ Assisted course coordination and grading · Global agribusiness strategy and operations · covered topics (international business environment · global agriculture suppy chain)
- **Teaching Material Development** — *2025*
  Data Science: Tools & Projects — Online Repository
  ◦ Practical examples & reusable code for quantitative policy analysis
  ◦ Methods: Python, Machine Learning, NLP, Geospatial Analysis, Causal Inference, LLMs, Dashboard Development
  ◦ Repository link: Data-Science-for-Policy-Research

Methods & Tools (Applied in Research)

Programming/Data/GIS/Version Control: Python · R · Stata · SQL · QGIS · Tableau · Git · GitHub
Applied Microeconometrics: DiD · event study · RD · IV · matching *(causal identification and policy evaluation)*
Machine Learning: random forest · gradient boosting · LASSO · basic deep learning *(risk monitoring and prediction)*
NLP & LLMs: BERTopic · named entity recognition · sentiment analysis · spaCy · Hugging Face · prompt engineering *(news analytics and document classification)*
Geospatial: geostatistical analysis · spatial econometrics · hotspot and cluster analysis *(mapping and spatial analysis)*

Presentations & Invited Talks

American Public Health Association (APHA) Annual Meeting and Expo (2026) · Thought Summit on DataScience to Build Resilience and Improve Humanitarian Response, Cornell CIDER (2026) · Food Assistance Strategic Dialogue, Johns Hopkins Bloomberg School of Public Health (2026) · AI + Health Care Idea Tank Pitch Competition, Hopkins Business of Health Initiative, Data Science and AI Institute, & Innovation Translation Council (2026) · American Agricultural Economics Association (AAEA) Annual Meeting (2025) · Center on Food Security and the Environment (Stanford) Seminar (2025) · European Association of Agricultural Economists (EAAE) Conference (2023) · Tata-Cornell Institute for Agriculture and Nutrition (TCI) Conference (2023) · American Agricultural Economics Association (AAEA) Annual Meeting (2022)

Other Activities

Professional Activities and Affiliations: Vice President, Graduate Student Organization, Department of Agricultural and Consumer Economics, UIUC (2020–2021) · Member, IPC Advanced Technologies and Artificial Intelligence (ATARI) Working Group (global initiative to enhance food security early warning leveraging AI tools) · Member, American Agricultural Economics Association (AAEA) (2022, 2025) · Member, European Association of Agricultural Economists (EAAE) (2023, 2025)
Public Engagement: Community engagement and humanitarian outreach, Champaign, Illinois (2020–2025):
Led donation drives and mobilized resources for programs such as the Shoebox Ministry, Soup Kitchen, and Ready Program · Organized emergency fundraising activities, including humanitarian relief efforts for UNHCR, engaging and coordinating over 100 members of the Korean community; focused on social responsibility, community solidarity, and inclusive action during times of need.

Honors & Awards

Finalist (Top 5 across JH schools), 2026 AI + Health Care Idea Tank Pitch Competition, Hopkins Business of Health Initiative, Data Science and AI Institute, & Innovation Translation Council (2026) · ACE Doctoral Fellowship, University of Illinois (Spring, Fall 2025) · Full Travel Grant, UNU–WIDER Development Conference (2025) · Travel Grant, American Agricultural Economics Association (AAEA) Annual Meeting (2025) · Departmental Travel Grant, ACE, University of Illinois (2025 Spring, 2025 Fall, 2022) · Banaba House Scholarship, Korea (2022–2024) · Cum Laude Graduate, Handong Global University (2015)

Languages

Korean (Native) · English (Fluent) · Chinese (Intermediate)

References

1. **Prof. Kathy Baylis**
   Professor and Vice-Chair
   Department of Geography
   University of California, Santa Barbara
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   Profile Page

2. **Prof. Erin Lentz**
   Associate Professor
   Lyndon B. Johnson School of Public Affairs
   University of Texas, Austin
   Email: erinclentz@utexas.edu
   Profile Page

3. **Prof. Hope Michelson**
   Professor
   Department of Agricultural and Consumer Economics
   University of Illinois, Urbana–Champaign
   Email: hopecm@illinois.edu
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